

Alpilla 600 MW Combined Cycle Power Plant Codes, Standards and Regulations (Appendix A.19), Permits and Authorities

163 register entries (100 agreed), 24 permits (24 agreed), 15 authorities. Agreed in TQ-033, TQ-IEC-022. Checks: 64 warning(s), 2 note(s).

1. Order of precedence (ER-01.07)

1. Turkish laws and regulations
2. Grid Code
3. Contract
4. Codes and standards (Appendix A.19) - where codes conflict the more stringent requirement applies
5. Good industry practice / project libraries

Edition: 'edition at Base Date' = the edition in force 28 days before the tender submission date; later editions apply only by variation. Responsible = parties whose design or work must comply (IEPC = Istanbul EPC, IEC = Imaginary Electric).

2. Laws, regulations and Grid Code

Code	Title	Edition	Authority	Applies to / obligations	Responsible	Status
Law 167	Groundwater Law (Yeralti Sulari Hakkinda Kanun)	1960, as amended	DSI	construction dewatering and groundwater wells	Istanbul EPC	agreed
Law 2863	Law on the Protection of Cultural and Natural Assets	1983, as amended	Museum Dir.	archaeological chance-find procedure during excavation	Istanbul EPC	agreed
Law 2872	Environmental Law (Cevre Kanunu)	1983, as amended	Ministry Env.	all environmental obligations; EIA conditions	Istanbul EPC, Imaginary Electric	agreed
Law 2918	Highway Traffic Law (Karayollari Trafik Kanunu)	1983, as amended	KGM	abnormal load transport permits	Istanbul EPC	agreed
Law 3194	Zoning Law (Imar Kanunu)	1985, as amended	Municipality	building permit and occupancy permit	Istanbul EPC, Imaginary Electric	agreed
Law 3621	Coastal Law (Kiyi Kanunu)	1990, as amended	Ministry Env.	intake and outfall pipelines and the temporary barge landing on the coastal strip	Istanbul EPC	agreed
Law 4646	Natural Gas Market Law (Dogal Gaz Piyasasi Kanunu)	2001, as amended	Gas network operator	gas connection, metering and station approval by the network operator	Istanbul EPC	agreed
Law 6331	Occupational Health and Safety Law (Is Sagligi ve Guvenligi Kanunu)	2012	MoLSS	design and construction safety, risk assessment	Istanbul EPC, Imaginary Electric	agreed
Law 6446	Electricity Market Law (Elektrik Piyasasi Kanunu)	2013	EMRA	generation licence, connection and system use	Owner	agreed
Law 7381	Nuclear Regulatory Law (Nukleer Duzenleme Kanunu)	2022	NDK	licensing of radiation sources for radiographic testing	Istanbul EPC, Imaginary Electric	agreed
Construction OHS Regulation	Regulation on OHS in Construction Works (Yapi Islerinde ISG Yonetmeligi)	2013	MoLSS	site safety plan, coordination between contractors	Istanbul EPC, Imaginary Electric	agreed
EIA Regulation	Environmental Impact Assessment Regulation (CED Yonetmeligi)	2022	Ministry Env.	conditions of the positive EIA decision (ALP-OWN-ENV-001); changes to the project need an EIA review	Istanbul EPC, Imaginary Electric	agreed

Code	Title	Edition	Authority	Applies to / obligations	Responsible	Status
Earthing Regulation	Regulation on Earthing in Electrical Installations (Elektrik Tesislerinde Topraklamalar Yonetmeligi)	2001	MENR	earthing system design and verification	Istanbul EPC, Imaginary Electric	agreed
Env. Permit Regulation	Environmental Permit and Licence Regulation (Cevre Izin ve Lisans Yonetmeligi)	2014	Ministry Env.	temporary activity certificate and environmental permit (air emissions, discharges)	Istanbul EPC, Imaginary Electric	agreed
Explosive Atmospheres Regulation	Regulation on Protection of Workers from Explosive Atmospheres	2013	MoLSS	explosion protection document for the gas station, GT package, LDO, H2	Istanbul EPC, Imaginary Electric	agreed
Fire Regulation	Regulation on Fire Protection of Buildings (Binalarin Yangindan Korunmasi Hakkinda Yonetmelik)	2007, as amended	Municipality	fire protection design approval of buildings	Istanbul EPC, Imaginary Electric	agreed
Heavy Current Regulation	Regulation on Electrical Heavy Current Installations (Elektrik Kuvvetli Akim Tesisleri Yonetmeligi)	2000	MENR	HV/MV installations, clearances	Istanbul EPC, Imaginary Electric	agreed
LCP Regulation	Large Combustion Plants Regulation (Buyuk Yakma Tesisleri Yonetmeligi)	2010	Ministry Env.	emission limit values of gas turbines > 50 MWth; the stricter EIA limits (ER-14.02) govern	Istanbul EPC, Imaginary Electric	agreed
Major Accident Regulation	Regulation on Prevention and Mitigation of Major Industrial Accidents (Seveso III)	2019	Ministry Env.	LDO storage about 5,850 t (2 x 3,500 m3) exceeds the 2,500 t lower-tier threshold for petroleum products: notification and Major Accident Prevention Policy; aqueous ammonia and hydrogen inventories added by the aggregation rule	Istanbul EPC, Imaginary Electric	agreed
Noise Regulation	Environmental Noise Control Regulation (Cevresel Gurultunun Kontrolu Yonetmeligi)	edition at Base Date	Ministry Env.	noise at the receptors (45/55 dB(A)), construction noise assessment	Istanbul EPC, Imaginary Electric	agreed
SKHKKY	Industrial Air Pollution Control Regulation (Sanayi Kaynakli Hava Kirliliginin Kontrolu Yonetmeligi)	2009	Ministry Env.	stack height and dispersion, emission measurement	Istanbul EPC, Imaginary Electric	agreed
Soil Regulation	Soil Pollution Control and Point-Source Contaminated Sites Regulation	2010	Ministry Env.	baseline and closure soil condition, spill prevention	Istanbul EPC	agreed
TBDY 2018	Turkish Building Earthquake Code (Turkiye Bina Deprem Yonetmeligi)	2018	Municipality	seismic design of buildings and structures; equipment anchorage	Istanbul EPC, Imaginary Electric	agreed
WPCR	Water Pollution Control Regulation (Su Kirliligi Kontrolu Yonetmeligi)	2004	Ministry Env.	sea discharge limits of process, sanitary and cooling water effluents	Istanbul EPC	agreed
Waste Regulation	Waste Management Regulation (Atik Yonetimi Yonetmeligi)	2015	Ministry Env.	construction and operation waste, hazardous waste	Istanbul EPC	agreed
Work Equipment Regulation	Regulation on Health and Safety in the Use of Work Equipment	2013	MoLSS	first and periodic inspection of lifting and pressure equipment before use	Istanbul EPC, Imaginary Electric	agreed
ATEX 2014/34/EU	ATEX Equipment Directive, transposed as the regulation on equipment for potentially explosive atmospheres	2014/34/EU	MoIT	equipment in hazardous areas	Istanbul EPC, Imaginary Electric	agreed
Machinery Directive 2006/42/EC	Machinery Directive, transposed as the Machinery Safety Regulation (Makine Emniyeti Yonetmeligi)	2006/42/EC	MoIT	CE marking and declarations of conformity of machinery packages (GT, ST, pumps, cranes)	Istanbul EPC, Imaginary Electric	agreed
PED 2014/68/EU	Pressure Equipment Directive, transposed as the Pressure Equipment Regulation (Basincli Ekipmanlar Yonetmeligi)	2014/68/EU	MoIT	conformity assessment of pressure equipment by a notified body (HRSG, vessels, piping)	Istanbul EPC, Imaginary Electric	agreed
Grid Code	Electricity Grid Regulation (Elektrik Sebeke Yonetmeligi) - TSO Grid Code	edition at Base Date	TSO	generator performance, protection, frequency and voltage control, compliance tests	Istanbul EPC, Imaginary Electric	agreed
Turkish regulations on noise at work and on chemical agents at work	Calisanlarin Gurultu ile Ilgili Risklerden Korunmasi Hakkinda Yonetmelik; Kimyasal Maddelerle Calisirken Saglik ve Guvenlik Onlemleri Hakkinda Yonetmelik	edition at Base Date	-	Occupational exposure limits used in HS-DBR-0001 section 6	Istanbul EPC	proposed

3. Codes and standards

Code	Title	Edition	Applies to	ER	Responsible	Deviation / clarification	Status
TS 500	Requirements for design and construction of reinforced concrete structures	edition at Base Date	reinforced concrete (with EN 1992)	ER-13.04	Istanbul EPC	-	agreed
TS EN 590	Automotive fuels - Diesel - Requirements and test methods (basis of the LDO specification)	edition at Base Date	LDO delivered by the Owner's supplier (TP-L1)	ER-03.07	Owner	-	agreed
ACI 351.3R	Report on Foundations for Dynamic Equipment	edition at Base Date	turbine-generator foundations (IEC load data, IF-43)	ER-13.01	Istanbul EPC	-	agreed
API 650	Welded Tanks for Oil Storage	edition at Base Date	LDO and water storage tanks	ER-03.08	Istanbul EPC	-	agreed

Code	Title	Edition	Applies to	ER	Responsible	Deviation / clarification	Status
ASME B16.25	Buttwelding Ends	edition at Base Date	weld-end terminal points	-	Istanbul EPC, Imaginary Electric	-	agreed
ASME B16.34	Valves - Flanged, Threaded, and Welding End	edition at Base Date	valves	-	Istanbul EPC, Imaginary Electric	-	agreed
ASME B16.5	Pipe Flanges and Flanged Fittings NPS 1/2 through NPS 24	edition at Base Date	flanged terminal points	-	Istanbul EPC, Imaginary Electric	-	agreed
ASME B31.1	Power Piping	edition at Base Date	steam, feedwater, condensate and utility piping	-	Istanbul EPC	-	agreed
ASME B31.3	Process Piping	edition at Base Date	chemical, ammonia and LDO piping	-	Istanbul EPC	-	agreed
ASME B31.8	Gas Transmission and Distribution Piping Systems	edition at Base Date	gas receiving station and plant gas piping	ER-18.03	Istanbul EPC	-	agreed
ASME B36.10M	Welded and Seamless Wrought Steel Pipe	-	pipe dimensions and wall thicknesses (pipe_size records)	-	Istanbul EPC, Imaginary Electric	-	agreed
ASME PTC 12.2	Performance Test Code on Steam Surface Condensers	edition at Base Date	condenser performance test	ER-07.08	Imaginary Electric	-	agreed
ASME PTC 22	Performance Test Code on Gas Turbines	edition at Base Date	GT performance tests	-	Imaginary Electric	-	agreed
ASME PTC 4.4	Performance Test Code on Gas Turbine Heat Recovery Steam Generators	edition at Base Date	HRSG performance tests	-	Imaginary Electric	-	agreed
ASME PTC 46	Performance Test Code on Overall Plant Performance	edition at Base Date	plant performance tests and correction curves	ER-16.04, ER-12.05	Istanbul EPC, Imaginary Electric	-	agreed
ASME PTC 6	Performance Test Code on Steam Turbines	edition at Base Date	ST performance tests (flow section in the main steam line)	-	Istanbul EPC, Imaginary Electric	-	agreed
ASME Section I	ASME BPVC Section I - Rules for Construction of Power Boilers	edition at Base Date	HRSG pressure parts, boiler external piping	ER-06.02	Istanbul EPC, Imaginary Electric	Selected for the HRSG (IEC); PED module G by a notified body; boiler external piping by the EPC to ASME Section I / B31.1.	agreed
ASME Section IX	ASME BPVC Section IX - Welding, Brazing and Fusing Qualifications	edition at Base Date	welding procedure and welder qualification (alternative EN ISO 15614)	ER-15.05	Istanbul EPC, Imaginary Electric	-	agreed
ASME Section VIII Div. 1	ASME BPVC Section VIII Division 1 - Pressure Vessels	edition at Base Date	BOP pressure vessels (flash tanks, air receivers, deaeration)	-	Istanbul EPC	-	agreed
EN 10204	Metallic products - Types of inspection documents	2004	material certificates 3.1 / 3.2	ER-15.04	Istanbul EPC, Imaginary Electric	-	agreed
EN 12186	Gas infrastructure - Gas pressure regulating stations for transmission and distribution	edition at Base Date	gas pressure regulation and safety devices (slam-shut, relief)	ER-03.04	Istanbul EPC	-	agreed
EN 12952	Water-tube boilers and auxiliary installations (all parts)	edition at Base Date	HRSG - alternative to ASME Section I (ER-06.02); EN 12952-3 fatigue assessment for cyclic operation (ER-06.05)	ER-06.02, ER-06.05	Imaginary Electric	HRSG designed to ASME Section I with PED conformity by a notified body (module G); EN 12952-3 applied only for the low-cycle fatigue assessment of the thick-walled parts (HP drum, SH/RH outlet headers) per ER-06.05.	agreed
EN 13001	Cranes - General design (all parts)	edition at Base Date	turbine hall EOT crane and hoists	ER-10.07	Istanbul EPC	-	agreed
EN 14181	Stationary source emissions - Quality assurance of automated measuring systems	edition at Base Date	CEMS QAL1/QAL2/QAL3/AST	ER-12.04	Istanbul EPC	-	agreed
EN 15259	Air quality - Measurement of stationary source emissions - Measurement sections and sites	edition at Base Date	stack measurement ports and platforms	ER-06.06	Istanbul EPC	-	agreed
EN 15267-3	Air quality - Certification of automated measuring systems - Part 3	edition at Base Date	certified CEMS analysers	ER-12.04	Istanbul EPC	-	agreed
EN 1822	High efficiency air filters (EPA, HEPA and ULPA)	edition at Base Date	GT inlet air final filter class	ER-05.02	Imaginary Electric	-	agreed

Code	Title	Edition	Applies to	ER	Responsible	Deviation / clarification	Status
EN 1990	Eurocode - Basis of structural design (with EN 1991 actions)	edition at Base Date	structural design basis and actions (wind, snow, thermal)	-	Istanbul EPC	-	agreed
EN 1992	Eurocode 2 - Design of concrete structures	edition at Base Date	concrete structures and foundations	ER-13.04	Istanbul EPC	-	agreed
EN 1993	Eurocode 3 - Design of steel structures	edition at Base Date	steel structures	ER-13.03	Istanbul EPC	-	agreed
EN ISO 1461	Hot dip galvanized coatings on fabricated iron and steel articles	edition at Base Date	galvanizing of structural steel	ER-13.03	Istanbul EPC	-	agreed
EN ISO 15614	Specification and qualification of welding procedures for metallic materials - Welding procedure test	edition at Base Date	welding procedure qualification (alternative ASME IX)	ER-15.05	Istanbul EPC, Imaginary Electric	-	agreed
HEI 2629	HEI Standards for Steam Surface Condensers	12th edition	condenser thermal and mechanical design	ER-07.07, ER-07.08, ER-07.09, ER-07.10	Imaginary Electric	-	agreed
IAPWS TGD	IAPWS Technical Guidance Documents on cycle chemistry (HRSG, AVT, instrumentation)	edition at Base Date	water/steam chemistry of the HRSG plant	ER-09.03	Istanbul EPC, Imaginary Electric	-	agreed
IAPWS-IF97	Industrial Formulation 1997 for the Thermodynamic Properties of Water and Steam	1997	water/steam properties in heat balances and tests	-	Istanbul EPC, Imaginary Electric	-	agreed
IEC 60034-1	Rotating electrical machines - Part 1: Rating and performance	edition at Base Date	generators; also -3 for turbine-type generators	ER-11.02, ER-04.08	Imaginary Electric	-	agreed
IEC 60045-1	Steam turbines - Part 1: Specifications	edition at Base Date	steam turbine; rated and design conditions	-	Imaginary Electric	-	agreed
IEC 60076	Power transformers (all parts)	edition at Base Date	GSU, UAT and auxiliary transformers	ER-11.03	Istanbul EPC, Imaginary Electric	-	agreed
IEC 60079-10-1	Explosive atmospheres - Classification of areas - Explosive gas atmospheres	edition at Base Date	hazardous area classification	ER-14.06	Istanbul EPC, Imaginary Electric	-	agreed
IEC 60364	Low-voltage electrical installations (all parts)	edition at Base Date	LV installations	-	Istanbul EPC	-	agreed
IEC 60815	Selection and dimensioning of high-voltage insulators for polluted conditions	edition at Base Date	creepage distances of outdoor insulation (salt air)	ER-02.10	Istanbul EPC, Imaginary Electric	-	agreed
IEC 60909	Short-circuit currents in three-phase AC systems	edition at Base Date	short-circuit studies (TP-E1, IF-34..IF-38)	-	Istanbul EPC, Imaginary Electric	-	agreed
IEC 61511	Functional safety - Safety instrumented systems for the process industry sector	edition at Base Date	SIS incl. burner management and HRSG protection	ER-12.03	Istanbul EPC, Imaginary Electric	-	agreed
IEC 61850	Communication networks and systems for power utility automation	edition at Base Date	substation automation and protection interface (IF-41)	-	Istanbul EPC, Imaginary Electric	-	agreed
IEC 61936-1	Power installations exceeding 1 kV AC - Part 1: Common rules	edition at Base Date	HV/MV installations, transformer fire separation	-	Istanbul EPC	-	agreed
IEC 62271-203	High-voltage switchgear - Gas-insulated metal-enclosed switchgear for rated voltages above 52 kV	edition at Base Date	380 kV GIS	ER-11.01	Istanbul EPC	-	agreed
IEC 62305	Protection against lightning (all parts)	edition at Base Date	lightning protection and risk assessment	ER-02.12, ER-11.07	Istanbul EPC	-	agreed
IEC 62443	Security for industrial automation and control systems (all parts)	edition at Base Date	OT cybersecurity SL2 (DCS, TCS, zone/conduit at IF-40)	ER-05.06, ER-12.07	Istanbul EPC, Imaginary Electric	IEC TCS certified to IEC 62443-4-1 / -4-2 components, system SL2 per IEC 62443-3-3; zone/conduit boundary and firewall at IF-40 by the EPC.	agreed
IEC 62770	Fluids for electrotechnical applications - Unused natural esters for transformers	edition at Base Date	GSU insulating liquid (natural ester)	-	Imaginary Electric	-	agreed
IEC/IEEE 62271-37-013	High-voltage switchgear - Alternating-current generator circuit-breakers	edition at Base Date	generator circuit breakers	ER-11.02	Imaginary Electric	-	agreed

Code	Title	Edition	Applies to	ER	Responsible	Deviation / clarification	Status
IEEE 80	Guide for Safety in AC Substation Grounding	edition at Base Date	earthing grid	ER-11.07	Istanbul EPC	-	agreed
IEEE C37.102	Guide for AC generator protection	edition at Base Date	generator protection functions	-	Imaginary Electric	-	agreed
ISO 10494	Gas turbines and gas turbine sets - Measurement of emitted airborne noise	edition at Base Date	GT noise data and tests	ER-14.03	Imaginary Electric	-	agreed
ISO 12944-2	Paints and varnishes - Corrosion protection of steel structures by protective paint systems - Part 2: Classification of environments	edition at Base Date	corrosivity category of outdoor equipment (coastal)	ER-02.08	Istanbul EPC, Imaginary Electric	-	agreed
ISO 14001	Environmental management systems - Requirements	2015	environmental management during construction and commissioning (ESMP)	-	Istanbul EPC, Imaginary Electric	-	agreed
ISO 20816-2	Mechanical vibration - Measurement and evaluation of machine vibration - Part 2: Land-based gas turbines, steam turbines and generators above 40 MW with fluid-film bearings	edition at Base Date	vibration limits of the GT, ST and generators (IEC)	-	Imaginary Electric	-	agreed
ISO 21789	Gas turbine applications - Safety	edition at Base Date	GT package	-	Imaginary Electric	-	agreed
ISO 3977-3	Gas turbines - Procurement - Part 3: Design requirements	edition at Base Date	GT package	-	Imaginary Electric	-	agreed
ISO 45001	Occupational health and safety management systems	2018	HSE management system	ER-14.05	Istanbul EPC, Imaginary Electric	-	agreed
ISO 6976	Natural gas - Calculation of calorific values, density, relative density and Wobbe indices	2016	fuel LHV in heat balances and performance tests	-	Istanbul EPC, Imaginary Electric	-	agreed
ISO 9001	Quality management systems - Requirements	2015	quality management of the Contractor and major suppliers	ER-15.01	Istanbul EPC, Imaginary Electric	-	agreed
NFPA 11	Standard for Low-, Medium-, and High-Expansion Foam	edition at Base Date	LDO tank foam protection	ER-03.08	Istanbul EPC	-	agreed
NFPA 12	Standard on Carbon Dioxide Extinguishing Systems	edition at Base Date	GT enclosure CO2 system	ER-10.05	Imaginary Electric	CO2 total flooding for the GT enclosure (ER-10.05 allows water mist or CO2); no water mist.	agreed
NFPA 20	Standard for the Installation of Stationary Pumps for Fire Protection	edition at Base Date	fire pumps	ER-10.05	Istanbul EPC	-	agreed
NFPA 30	Flammable and Combustible Liquids Code	edition at Base Date	LDO unloading, storage and bunds	ER-03.08	Istanbul EPC	-	agreed
NFPA 72	National Fire Alarm and Signaling Code	edition at Base Date	fire detection and alarm (IF-33)	ER-10.05	Istanbul EPC	-	agreed
NFPA 85	Boiler and Combustion Systems Hazards Code	edition at Base Date	HRSG purge and combustion turbine exhaust systems	-	Imaginary Electric	-	agreed
NFPA 850	Recommended Practice for Fire Protection for Electric Generating Plants and HVDC Converter Stations	edition at Base Date	plant fire protection concept	ER-10.05	Istanbul EPC, Imaginary Electric	-	agreed
VGB-S-010	Feed water, boiler water and steam quality for power plants	edition at Base Date	cycle chemistry limits	ER-09.03	Istanbul EPC, Imaginary Electric	-	agreed
AVEVA class library 1.0.1	AVEVA engineering class library (classes, attributes, units, value lists) - project tag classification standard	modelUpdated AtVersion 4.0.0.0	engineering classes and attributes of the project database	-	Istanbul EPC	-	agreed
TS 708	Steel for the reinforcement of concrete - reinforcing steel	edition at Base Date	B500C ribbed bars per TBDY 2018	-	Istanbul EPC	-	proposed
TS EN 197-1	Cement - composition, specifications and conformity criteria for common cements	edition at Base Date	Cement types for XS / XA exposure	-	Istanbul EPC	-	proposed
API 682 / ISO 21049	Pumps - Shaft sealing systems for centrifugal and rotary pumps	edition at Base Date	Mechanical seals for pumps in LDO, ammonia and hazardous service	-	Istanbul EPC	-	proposed
ASHRAE 62.1 / 55 / 90.1	Ventilation for acceptable indoor air quality; thermal comfort; energy standard for buildings	edition at Base Date	HVAC design method and ventilation rates (HV-DBR-0001)	-	Istanbul EPC	-	proposed

Code	Title	Edition	Applies to	ER	Responsible	Deviation / clarification	Status
ASME B16.47	Large Diameter Steel Flanges NPS 26 through NPS 60	edition at Base Date	Flanges of large diameter piping (CW, condensate, bypass steam)	-	Istanbul EPC	-	proposed
ASME PTC 19.3 TW	Thermowells	edition at Base Date	Thermowell wake frequency check (IC-DBR s.10)	-	Istanbul EPC	-	proposed
ASTM A923 / ASTM G48	Detecting detrimental intermetallic phase in duplex stainless steels; pitting and crevice corrosion resistance tests	edition at Base Date	Super duplex acceptance tests in the materials report	-	Istanbul EPC	-	proposed
EEMUA 191 / IEC 62682 / ANSI-ISA-18.2	Alarm systems - design, management and procurement	edition at Base Date	Alarm management (IC-DBR s.12); named in TQ-026	-	Istanbul EPC	-	proposed
EN 10025	Hot rolled products of structural steels	edition at Base Date	Steel grades S355J2 / S275JR	-	Istanbul EPC	-	proposed
EN 1090-2	Execution of steel structures and aluminium structures - technical requirements	edition at Base Date	Execution class EXC3 for power block steelwork (ER-13.03)	-	Istanbul EPC	-	proposed
EN 12464-1 / EN 12464-2 / EN 1838	Lighting of work places (indoor, outdoor) and emergency lighting	edition at Base Date	Illuminance criteria (EL-DBR s.14)	-	Istanbul EPC	-	proposed
EN 13445	Unfired pressure vessels (all parts)	edition at Base Date	Pressure vessel design code alternative to ASME VIII	-	Istanbul EPC	-	proposed
EN 16798-1	Energy performance of buildings - ventilation for buildings - indoor environmental input parameters	edition at Base Date	Indoor conditions and ventilation rates	-	Istanbul EPC	-	proposed
EN 1991-1-4 (and -1-1, -1-3, -3)	Eurocode 1 - Actions on structures: wind, imposed, snow, cranes	edition at Base Date	Wind, snow, imposed and crane loads; only EN 1990 is registered as 'with EN 1991 actions'	-	Istanbul EPC	-	proposed
EN 1997-1	Eurocode 7 - Geotechnical design - General rules	edition at Base Date	Pile and shallow foundation design, bearing capacity, settlement (CV-DBR-0001 sections 4, 13)	-	Istanbul EPC	-	proposed
EN 206	Concrete - Specification, performance, production and conformity	edition at Base Date	Exposure classes XS / XA, cement types and w/c limits for the marine environment (ER-13.04)	-	Istanbul EPC	-	proposed
EN 50272-2	Safety requirements for secondary batteries - stationary batteries	edition at Base Date	Battery room ventilation (EL-DBR s.9, s.13)	-	Istanbul EPC	-	proposed
EN ISO 16890	Air filters for general ventilation	edition at Base Date	Filter classes G4 / F7 / F9 for salt and dust filtration	-	Istanbul EPC	-	proposed
IEC 60034-30-1	Rotating electrical machines - efficiency classes (IE code)	edition at Base Date	Motor efficiency (EL-DBR s.8)	-	Istanbul EPC	-	proposed
IEC 60068-3-3 / IEEE 344	Seismic test methods / seismic qualification of equipment	edition at Base Date	Equipment qualification for PGA 0.45 g (EL-DBR s.3)	-	Istanbul EPC	-	proposed
IEC 60071-1/-2	Insulation co-ordination	edition at Base Date	Insulation levels and study EL-STU-0005 (EL-DBR s.15-16)	-	Istanbul EPC	-	proposed
IEC 60079-0, IEC 60079-14, IEC 60079-17	Explosive atmospheres: equipment general requirements, installation, inspection	edition at Base Date	Hazardous area equipment (EL-DBR s.13)	-	Istanbul EPC	-	proposed
IEC 60287	Electric cables - calculation of the current rating	edition at Base Date	Cable derating (EL-DBR s.10)	-	Istanbul EPC	-	proposed
IEC 60332-3, IEC 60754, IEC 61034, IEC 60331	Cable fire tests: flame spread, halogen content, smoke density, circuit integrity	edition at Base Date	ER-11.08 fire performance (EL-DBR s.10)	-	Istanbul EPC	-	proposed
IEC 60502	Power cables with extruded insulation for 1 kV to 30 kV	edition at Base Date	Cable criteria (EL-DBR s.10)	-	Istanbul EPC	-	proposed
IEC 60529	Degrees of protection provided by enclosures (IP code)	edition at Base Date	Instrument and cabinet ingress protection (IC-DBR s.10)	-	Istanbul EPC	-	proposed
IEC 60534 (parts 2-1, 4, 8-3, 8-4)	Industrial-process control valves: sizing, leakage, noise	edition at Base Date	Control valve criteria (IC-DBR s.11)	-	Istanbul EPC	-	proposed
IEC 60751 / IEC 60584	Industrial platinum resistance thermometers / thermocouples	edition at Base Date	Temperature sensors (IC-DBR s.9)	-	Istanbul EPC	-	proposed
IEC 60870-5-104	Telecontrol equipment - network access	edition at Base Date	RTU to dispatch centre (EL-DBR s.11)	-	Istanbul EPC	-	proposed
IEC 60896-21/-22	Stationary lead-acid batteries - valve regulated	edition at Base Date	Battery selection (EL-DBR s.9)	-	Istanbul EPC	-	proposed

Code	Title	Edition	Applies to	ER	Responsible	Deviation / clarification	Status
IEC 61000-2-4 / IEC 61000-6-2 / IEC 61000-6-4	EMC: compatibility levels in industrial plants, immunity and emission standards	edition at Base Date	Harmonics limit (EL-DBR s.8, s.16) and DCS EMC (IC-DBR s.9)	-	Istanbul EPC	-	proposed
IEC 61439	Low-voltage switchgear and controlgear assemblies	edition at Base Date	LV boards and MCCs (EL-DBR s.4)	-	Istanbul EPC	-	proposed
IEC 61508	Functional safety of E/E/PE safety-related systems	edition at Base Date	SIL certification of SIS devices (IC-DBR s.8, s.10)	-	Istanbul EPC	-	proposed
IEC 61869	Instrument transformers	edition at Base Date	Metering and protection CT/VT (EL-DBR s.11)	-	Istanbul EPC	-	proposed
IEC 62271-200	AC metal-enclosed switchgear and controlgear for rated voltages above 1 kV up to and including 52 kV	edition at Base Date	MV switchgear type test and internal arc classification (EL-DBR s.4, s.11)	-	Istanbul EPC	-	proposed
IEC 62439-3	Industrial networks - PRP and HSR	edition at Base Date	Redundant IEC 61850 station bus (EL-DBR s.11)	-	Istanbul EPC	-	proposed
IEC 62541 (OPC UA)	OPC Unified Architecture	edition at Base Date	TCS to DCS integration (IC-DBR s.5)	-	Istanbul EPC	-	proposed
IEEE 1584	Guide for performing arc-flash hazard calculations	edition at Base Date	Arc-flash study in protection coordination (EL-DBR s.16)	-	Istanbul EPC	-	proposed
IEEE C37.238	IEEE Standard Profile for use of IEEE 1588 PTP in power system applications	edition at Base Date	Time synchronisation of relays (EL-DBR s.11, IC-DBR s.14)	-	Istanbul EPC	-	proposed
ISA 71.04	Environmental conditions for process measurement and control systems: airborne contaminants	edition at Base Date	Gaseous contaminant severity level for control rooms in the coastal atmosphere	-	Istanbul EPC	-	proposed
ISO 10816-7	Mechanical vibration - Evaluation of machine vibration by measurements on non-rotating parts - Part 7: Rotodynamic pumps for industrial applications	edition at Base Date	pump vibration acceptance (ME-SPC-0001 section 5)	-	Istanbul EPC	-	proposed
ISO 12944-1, -3 to -9	Paints and varnishes - Corrosion protection of steel structures by protective paint systems	edition at Base Date	Only ISO 12944-2 is registered; systems, design, application, inspection parts are cited in the painting specification	-	Istanbul EPC	-	proposed
ISO 13707 / API 617 / NEMA SM23	Allowable nozzle loads for rotating equipment	edition at Base Date	Nozzle load check in piping stress analysis	-	Istanbul EPC	-	proposed
ISO 13709 / API 610	Centrifugal pumps for petroleum, petrochemical and natural gas industries	edition at Base Date	Pump design standard cited in ME-SPC-0001 (only API 650 / ASME B31 etc. are registered)	-	Istanbul EPC	-	proposed
ISO 14692 / EN 1796 / AWWA M45	GRP piping design and installation	edition at Base Date	Seawater GRP piping design	-	Istanbul EPC	-	proposed
ISO 15848-1	Industrial valves - measurement of fugitive emissions	edition at Base Date	Stem emission of fuel gas valves (IC-DBR s.11)	-	Istanbul EPC	-	proposed
ISO 2063	Thermal spraying - Zinc, aluminium and their alloys	edition at Base Date	Thermally sprayed aluminium under insulation	-	Istanbul EPC	-	proposed
ISO 20816-3	Mechanical vibration - evaluation of machine vibration, industrial machines	edition at Base Date	BOP machine vibration monitoring (IC-DBR s.10)	-	Istanbul EPC	-	proposed
ISO 3864 / ISO 7010	Graphical symbols - Safety colours and safety signs	edition at Base Date	Safety colours and hazard marking	-	Istanbul EPC	-	proposed
ISO 5167	Measurement of fluid flow by differential pressure devices	edition at Base Date	Flow measurement (IC-DBR s.10)	-	Istanbul EPC	-	proposed
ISO 5199	Technical specifications for centrifugal pumps - Class II	edition at Base Date	Pump standard for water and utility services in ME-SPC-0001	-	Istanbul EPC	-	proposed
ISO 5801 / ISO 14694 / ISO 12759	Fans - performance, balance and vibration, efficiency classification	edition at Base Date	Fans in ME-SPC-0001	-	Istanbul EPC	-	proposed
ISO 8501-1 / ISO 8501-3	Preparation of steel substrates before application of paints - visual assessment of surface cleanliness	edition at Base Date	Sa 2.5 surface preparation	-	Istanbul EPC	-	proposed
ISO 8502 / ISO 8503 / ISO 19840	Surface cleanliness tests, surface profile and DFT measurement	edition at Base Date	Coating inspection methods	-	Istanbul EPC	-	proposed
ISO 8528-1/-5	Reciprocating IC engine driven AC generating sets	edition at Base Date	EDG rating and load-step performance (EL-DBR s.9)	-	Istanbul EPC	-	proposed
ISO 9612	Acoustics - determination of occupational noise exposure	edition at Base Date	Occupational noise assessment	-	Istanbul EPC	-	proposed

Code	Title	Edition	Applies to	ER	Responsible	Deviation / clarification	Status
ISO 9906	Rotodynamic pumps - Hydraulic performance acceptance tests, Grades 1, 2 and 3	edition at Base Date	Pump performance guarantees	-	Istanbul EPC	-	proposed
NAMUR NE43 / NE107	Signal level for failure information / self-monitoring and diagnosis of field devices	edition at Base Date	4-20 mA HART fault signals and device status (IC-DBR s.9)	-	Istanbul EPC	-	proposed
NFPA 101	Life Safety Code	edition at Base Date	Escape route criteria complementary to the Fire Regulation	-	Istanbul EPC	-	proposed
NFPA 15 / NFPA 24 / NFPA 13	Water spray fixed systems; private fire service mains; sprinkler systems	edition at Base Date	Transformer deluge, ring main and hydrants, building sprinklers	-	Istanbul EPC	-	proposed
NFPA 90A	Standard for the installation of air-conditioning and ventilating systems	edition at Base Date	Fire dampers, smoke shutdown of AHUs	-	Istanbul EPC	-	proposed
TEMA	Standards of the Tubular Exchanger Manufacturers Association	edition at Base Date	Shell-and-tube exchanger design	-	Istanbul EPC	-	proposed

4. Permits, licences and approvals

ID	Permit	Authority	Legal basis	Applicant	Deliverables (who provides)	Apply by / lead	Needed by / gates	Permit status
PMT-001	Positive EIA decision	Ministry Env.	Law 2872, EIA Regulation	Owner	-	- / - d	2026-11-02 all works	granted
PMT-002	Generation licence	EMRA	Law 6446	Owner	-	- / - d	2026-11-02 Contract effectiveness	granted
PMT-003	Grid connection agreement	TSO	Law 6446, Grid Code	Owner	-	- / - d	2026-11-02 grid design	granted
PMT-010	Workplace registration and construction site notification	SGK	Law 6331, Construction OHS Regulation	Istanbul EPC	EPC: site safety plan, OHS professionals	2026-11-02 / 10 d	2026-11-16 mobilisation	in preparation
PMT-025	Archaeological chance-find notifications	Museum Dir.	Law 2863	Istanbul EPC	EPC: chance-find procedure in the construction ESMP	- / - d	2026-11-16 excavations	planned
PMT-011	Construction noise assessment report	Ministry Env.	Noise Regulation, EIA Regulation	Istanbul EPC	EPC: acoustic report, working hours plan	2026-12-01 / 30 d	2027-01-15 piling	planned
PMT-022	Storm water discharge consent	DSI	WPCR	Owner	EPC: drainage design, oil interceptors	- / - d	2027-01-15 site drainage	granted
PMT-005	Building permit (yapi ruhsati)	Municipality	Law 3194, TBDY 2018, Fire Regulation	Owner	EPC: architectural, structural (TBDY 2018), mechanical and electrical projects signed by licensed Turkish engineers, geotechnical report, fire approval; IEC: equipment loads and foundation drawings	2026-11-16 / 60 d	2027-02-15 start of permanent foundations (piling)	in preparation
PMT-012	Construction dewatering permit	DSI	Law 167	Istanbul EPC	EPC: dewatering design and discharge plan	2026-11-16 / 60 d	2027-02-15 deep excavations	planned
PMT-023	Construction power connection	DISCO	Law 6446	Owner	EPC: load list, RMU design	2026-12-15 / 60 d	2027-03-15 construction power	applied
PMT-006	Coastal permit for intake and outfall pipelines	Ministry Env.	Law 3621, EIA Regulation	Owner	EPC: marine design, route survey, construction method statement	2026-11-02 / 180 d	2027-10-01 marine works (window 1 October - 30 April)	applied
PMT-007	Coastal permit for the temporary barge landing	Ministry Env.	Law 3621	Owner	EPC: landing design and removal plan by 2027-03-31	2027-04-01 / 180 d	2027-11-01 first heavy delivery by barge	planned
PMT-008	Harbour master approval for barge operations	MoTI / Harbour Master	Law 3621	Istanbul EPC	EPC: marine operations plan; IEC: heavy piece list (weights, dimensions)	2027-09-15 / 30 d	2027-11-01 first heavy delivery by barge	planned
PMT-009	Abnormal load road transport permits	KGM	Law 2918	Istanbul EPC	EPC: route survey; IEC: transport drawings	2027-09-01 / 30 d	2027-11-01 heavy haul to site	planned
PMT-013	Radiation source licence for radiographic testing	NDK	Law 7381	Istanbul EPC	NDT subcontractor licence, radiation protection plan	2027-09-15 / 60 d	2027-12-01 first site radiography	planned
PMT-004	System use agreement and energisation approval	TSO	Law 6446, Grid Code	Owner	EPC: protection settings, SCADA test report, single-line diagram; IEC: generator and GSU data, excitation and governor models	2028-03-15 / 90 d	2028-07-15 KD-010 back-feed	planned

ID	Permit	Authority	Legal basis	Applicant	Deliverables (who provides)	Apply by / lead	Needed by / gates	Permit status
PMT-016	First inspection of lifting and pressure equipment	MoLSS	Work Equipment Regulation, PED 2014/68/EU	Istanbul EPC	EPC and IEC: declarations of conformity, notified body certificates, load tests	2028-06-15 / 14 d	2028-08-01 use of equipment in commissioning	planned
PMT-014	Gas connection and receiving station approval	Gas network operator	Law 4646	Owner	EPC: station design (EN 12186 / ASME B31.8), metering, pressure protection, test records; IEC: GT gas demand and pressure requirements	2028-03-15 / 120 d	2028-08-15 gas-in (TP-G1)	planned
PMT-017	Emission data link registration (continuous emission monitoring centre)	Ministry Env.	Env. Permit Regulation, SKHKKY	Owner	EPC: CEMS certificates (EN 15267), QAL1, data link test	2028-07-15 / 45 d	2028-10-02 first fire (KD-020)	planned
PMT-018	Temporary activity certificate (environmental)	Ministry Env.	Env. Permit Regulation, LCP Regulation, WPCR	Owner	EPC: emission and discharge measurement plan, CEMS; IEC: GT emission data	2028-08-15 / 30 d	2028-10-02 first fire (KD-020)	planned
PMT-015	Major accident notification and Major Accident Prevention Policy	Ministry Env.	Major Accident Regulation	Owner	EPC: hazardous substance inventory, QRA inputs, MAPP draft; IEC: inventories in the power island (lube oil, H2, ammonia skid)	2028-08-15 / 60 d	2028-11-01 LDO first fill (TP-L1)	planned
PMT-019	Environmental permit (air emissions and discharges)	Ministry Env.	Env. Permit Regulation, LCP Regulation, WPCR	Owner	EPC: QAL2 report, accredited emission and discharge test reports (ER-16.05)	2029-03-01 / 90 d	2029-07-04 Taking-Over (KD-040)	planned
PMT-020	Occupancy permit (yapi kullanma izin belgesi)	Municipality	Law 3194, Fire Regulation	Owner	EPC: as-built drawings, fire system test certificates	2029-05-15 / 30 d	2029-07-04 Taking-Over (KD-040)	planned
PMT-021	Grid Code compliance approval (provisional and final)	TSO	Grid Code	Owner	EPC and IEC: test programme, validated models, test reports (ER-16.03)	2029-01-02 / 60 d	2029-07-04 commercial operation	planned

5. Authorities

ID	Authority	Regulations enforced	Permits
DISCO	Regional electricity distribution company (construction power)	-	PMT-023
DSI	General Directorate of State Hydraulic Works (surface and groundwater)	Law 167	PMT-012, PMT-022
EMRA	Energy Market Regulatory Authority (EPDK)	Law 6446	PMT-002
GASNO	Natural gas transmission network operator (connection and station approval)	Law 4646	PMT-014
KGM	General Directorate of Highways (abnormal load permits)	Law 2918	PMT-009
MENR	Ministry of Energy and Natural Resources	Heavy Current Regulation, Earthing Regulation	-
MOCT	Ministry of Culture and Tourism (regional museum directorate)	Law 2863	PMT-025
MOEUCC	Ministry of Environment, Urbanisation and Climate Change (and its Provincial Directorate)	Law 2872, EIA Regulation, Env. Permit Regulation, SKHKKY, LCP Regulation, WPCR, Noise Regulation, Waste Regulation, Soil Regulation, Major Accident Regulation, Law 3621	PMT-001, PMT-006, PMT-007, PMT-011, PMT-015, PMT-017, PMT-018, PMT-019
MOIT	Ministry of Industry and Technology (market surveillance of products)	PED 2014/68/EU, Machinery Directive 2006/42/EC, ATEX 2014/34/EU	-
MOLSS	Ministry of Labour and Social Security (OHS inspection)	Law 6331, Construction OHS Regulation, Work Equipment Regulation, Explosive Atmospheres Regulation	PMT-016
MOTI	Ministry of Transport and Infrastructure (Harbour Master, coastal structures)	-	PMT-008
MUNI	District municipality / provincial special administration (building, occupancy, fire approval, water)	Law 3194, Fire Regulation, TBDY 2018	PMT-005, PMT-020
NDK	Nuclear Regulatory Authority (radiation sources for NDT)	Law 7381	PMT-013
SGK	Social Security Institution (workplace registration)	-	PMT-010
TSO	National Transmission System Operator	Grid Code	PMT-003, PMT-004, PMT-021

6. Check results

Item	Check	Result	Detail
REF-EPCE-0049	agreement	WARN	API 682 / ISO 21049: status proposed
REF-EPCE-0038	agreement	WARN	ASHRAE 62.1 / 55 / 90.1: status proposed

Item	Check	Result	Detail
REF-EPCE-0057	agreement	WARN	ASME B16.47: status proposed
REF-EPCE-0028	agreement	WARN	ASME PTC 19.3 TW: status proposed
REF-EPCE-0061	agreement	WARN	ASTM A923 / ASTM G48: status proposed
REF-EPCE-0022	agreement	WARN	EEMUA 191 / IEC 62682 / ANSI-HA-18.2: status proposed
REF-EPCE-0035	agreement	WARN	EN 10025: status proposed
REF-EPCE-0034	agreement	WARN	EN 1090-2: status proposed
REF-EPCE-0016	agreement	WARN	EN 12464-1 / EN 12464-2 / EN 1838: status proposed
REF-EPCE-0050	agreement	WARN	EN 13445: status proposed
REF-EPCE-0039	agreement	WARN	EN 16798-1: status proposed
REF-EPCE-0033	agreement	WARN	EN 1991-1-4 (and -1-1, -1-3, -3): status proposed
REF-EPCE-0031	agreement	WARN	EN 1997-1: status proposed
REF-EPCE-0032	agreement	WARN	EN 206: status proposed
REF-EPCE-0013	agreement	WARN	EN 50272-2: status proposed
REF-EPCE-0040	agreement	WARN	EN ISO 16890: status proposed
REF-EPCE-0017	agreement	WARN	IEC 60034-30-1: status proposed
REF-EPCE-0019	agreement	WARN	IEC 60068-3-3 / IEEE 344: status proposed
REF-EPCE-0003	agreement	WARN	IEC 60071-1/-2: status proposed
REF-EPCE-0012	agreement	WARN	IEC 60079-0, IEC 60079-14, IEC 60079-17: status proposed
REF-EPCE-0006	agreement	WARN	IEC 60287: status proposed
REF-EPCE-0005	agreement	WARN	IEC 60332-3, IEC 60754, IEC 61034, IEC 60331: status proposed
REF-EPCE-0004	agreement	WARN	IEC 60502: status proposed
REF-EPCE-0024	agreement	WARN	IEC 60529: status proposed
REF-EPCE-0020	agreement	WARN	IEC 60534 (parts 2-1, 4, 8-3, 8-4): status proposed
REF-EPCE-0025	agreement	WARN	IEC 60751 / IEC 60584: status proposed
REF-EPCE-0009	agreement	WARN	IEC 60870-5-104: status proposed
REF-EPCE-0015	agreement	WARN	IEC 60896-21/-22: status proposed
REF-EPCE-0018	agreement	WARN	IEC 61000-2-4 / IEC 61000-6-2 / IEC 61000-6-4: status proposed
REF-EPCE-0002	agreement	WARN	IEC 61439: status proposed
REF-EPCE-0021	agreement	WARN	IEC 61508: status proposed
REF-EPCE-0008	agreement	WARN	IEC 61869: status proposed
REF-EPCE-0001	agreement	WARN	IEC 62271-200: status proposed
REF-EPCE-0007	agreement	WARN	IEC 62439-3: status proposed
REF-EPCE-0023	agreement	WARN	IEC 62541 (OPC UA): status proposed
REF-EPCE-0011	agreement	WARN	IEEE 1584: status proposed
REF-EPCE-0010	agreement	WARN	IEEE C37.238: status proposed
REF-EPCE-0043	agreement	WARN	ISA 71.04: status proposed
REF-EPCE-0063	agreement	WARN	ISO 10816-7: status proposed
REF-EPCE-0051	agreement	WARN	ISO 12944-1, -3 to -9: status proposed
REF-EPCE-0062	agreement	WARN	ISO 13707 / API 617 / NEMA SM23: status proposed
REF-EPCE-0047	agreement	WARN	ISO 13709 / API 610: status proposed
REF-EPCE-0056	agreement	WARN	ISO 14692 / EN 1796 / AWWA M45: status proposed
REF-EPCE-0029	agreement	WARN	ISO 15848-1: status proposed
REF-EPCE-0054	agreement	WARN	ISO 2063: status proposed
REF-EPCE-0030	agreement	WARN	ISO 20816-3: status proposed
REF-EPCE-0055	agreement	WARN	ISO 3864 / ISO 7010: status proposed
REF-EPCE-0027	agreement	WARN	ISO 5167: status proposed
REF-EPCE-0048	agreement	WARN	ISO 5199: status proposed
REF-EPCE-0059	agreement	WARN	ISO 5801 / ISO 14694 / ISO 12759: status proposed
REF-EPCE-0052	agreement	WARN	ISO 8501-1 / ISO 8501-3: status proposed
REF-EPCE-0053	agreement	WARN	ISO 8502 / ISO 8503 / ISO 19840: status proposed
REF-EPCE-0014	agreement	WARN	ISO 8528-1/-5: status proposed
REF-EPCE-0044	agreement	WARN	ISO 9612: status proposed
REF-EPCE-0060	agreement	WARN	ISO 9906: status proposed
REF-EPCE-0026	agreement	WARN	NAMUR NE43 / NE107: status proposed
REF-EPCE-0046	agreement	WARN	NFPA 101: status proposed
REF-EPCE-0042	agreement	WARN	NFPA 15 / NFPA 24 / NFPA 13: status proposed
REF-EPCE-0041	agreement	WARN	NFPA 90A: status proposed
REF-EPCE-0058	agreement	WARN	TEMA: status proposed

Item	Check	Result	Detail
REF-EPCE-0036	agreement	WARN	TS 708: status proposed
REF-EPCE-0037	agreement	WARN	TS EN 197-1: status proposed
REF-EPCE-0045	authority	WARN	Turkish regulations on noise at work and on chemical agents at work: regulation without enforcing authority
REF-EPCE-0045	agreement	WARN	Turkish regulations on noise at work and on chemical agents at work: status proposed
PMT-010	timing	INFO	apply 2026-11-02 + 10 d = 2026-11-12; needed 2026-11-16: float 4 d
PMT-025	timing	INFO	no lead time: procedural, needed from 2026-11-16